ARTIFICIAL INTELLIGENCE

HSU

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Random Forest (RF)

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- Is RF related to DT?

→ YES

- A **Random Forest** is an **ensemble machine learning method** that builds upon **decision trees** to improve accuracy and reduce overfitting.

- Think of it as a **"forest" of many decision trees**, where each tree gives a "vote," and the majority vote (or average, for regression) becomes the final prediction.

What is a Random Forest?

1. Bagging (Bootstrap Aggregating/Sampling):
   - take random subsets of data points from the training set to create N smaller data sets
   - fit a decision tree on each subset
   - every tree sees a slightly different version of the dataset.
   - results in low variance.

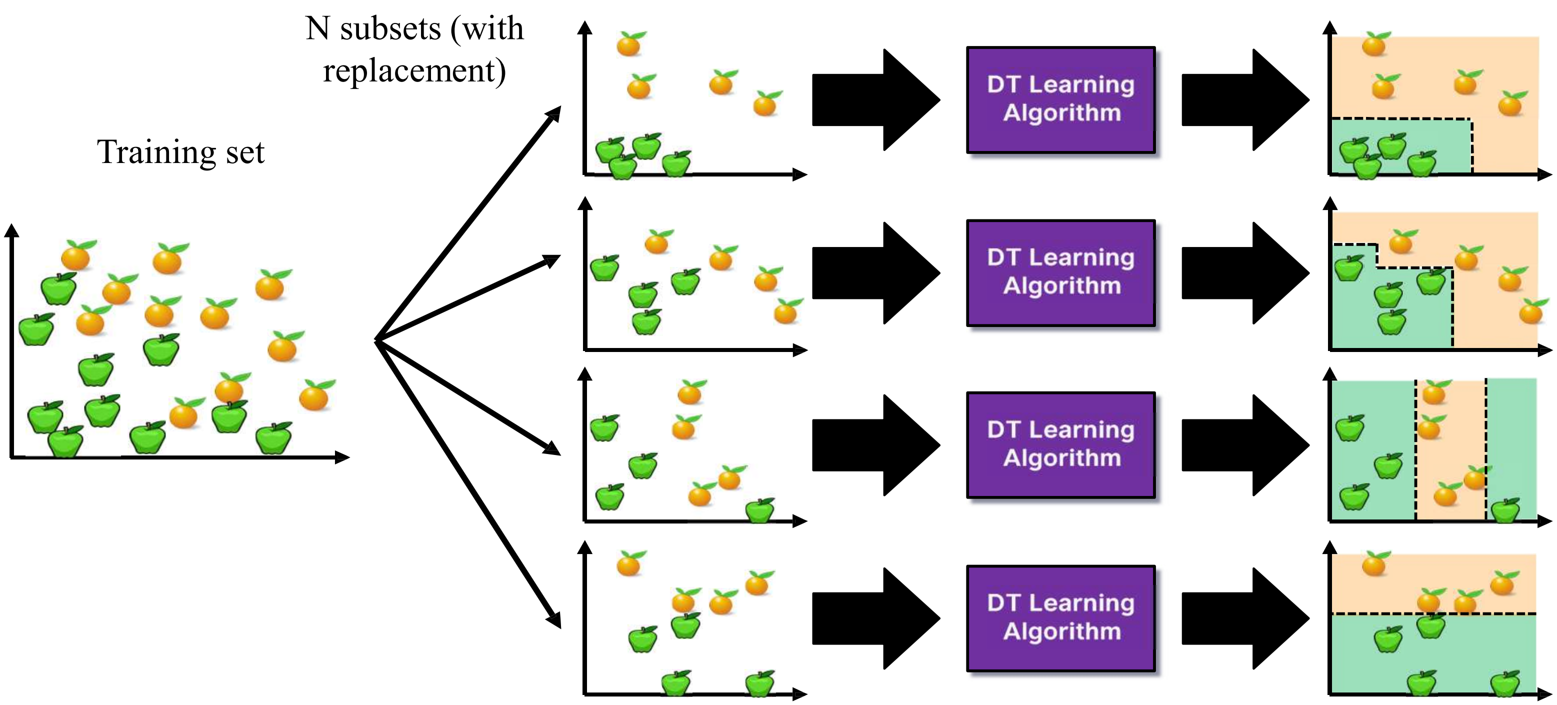
2. Random Subspace Method (Feature Selection)
   - Fit N different decision trees by constraining each one to operate on a random subset of features.
   - This adds extra randomness, making trees less correlated and improving generalization

Aggregation (Voting/Averaging)

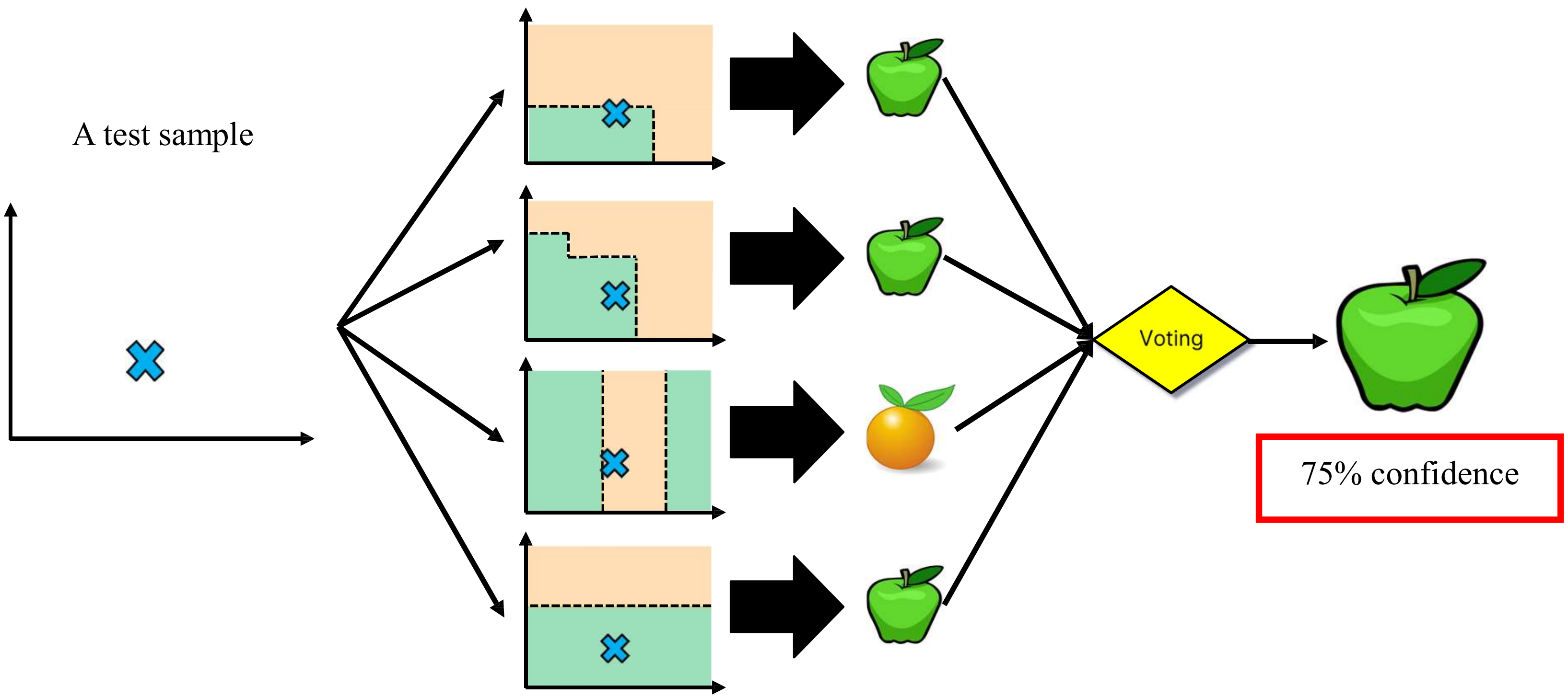
- **Classification:** Each tree votes for a class → the majority class is the prediction.

- **Regression:** Each tree outputs a number → the average is the prediction.

Bagging at training time



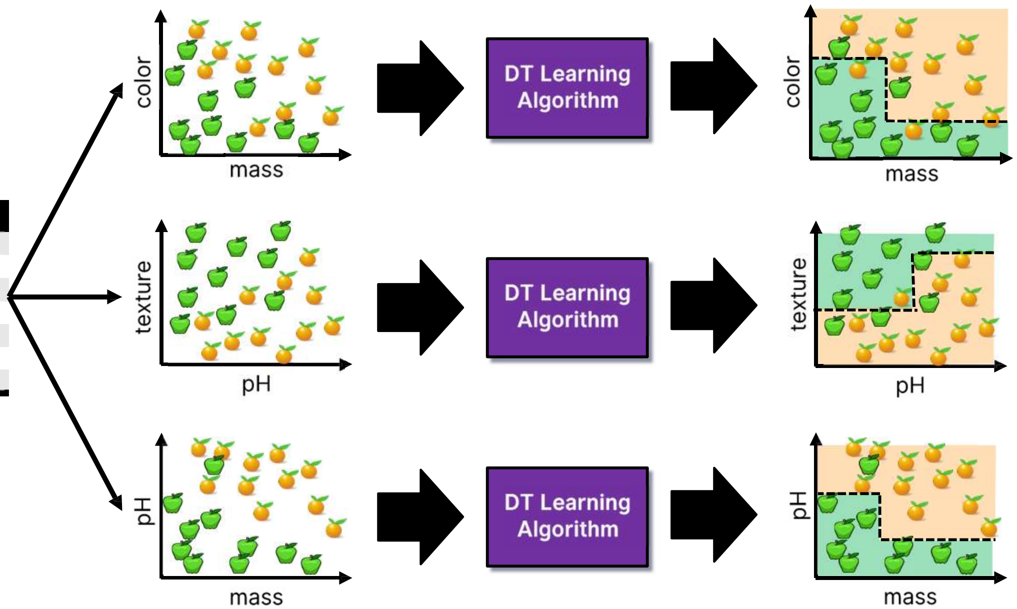
Bagging at inference time



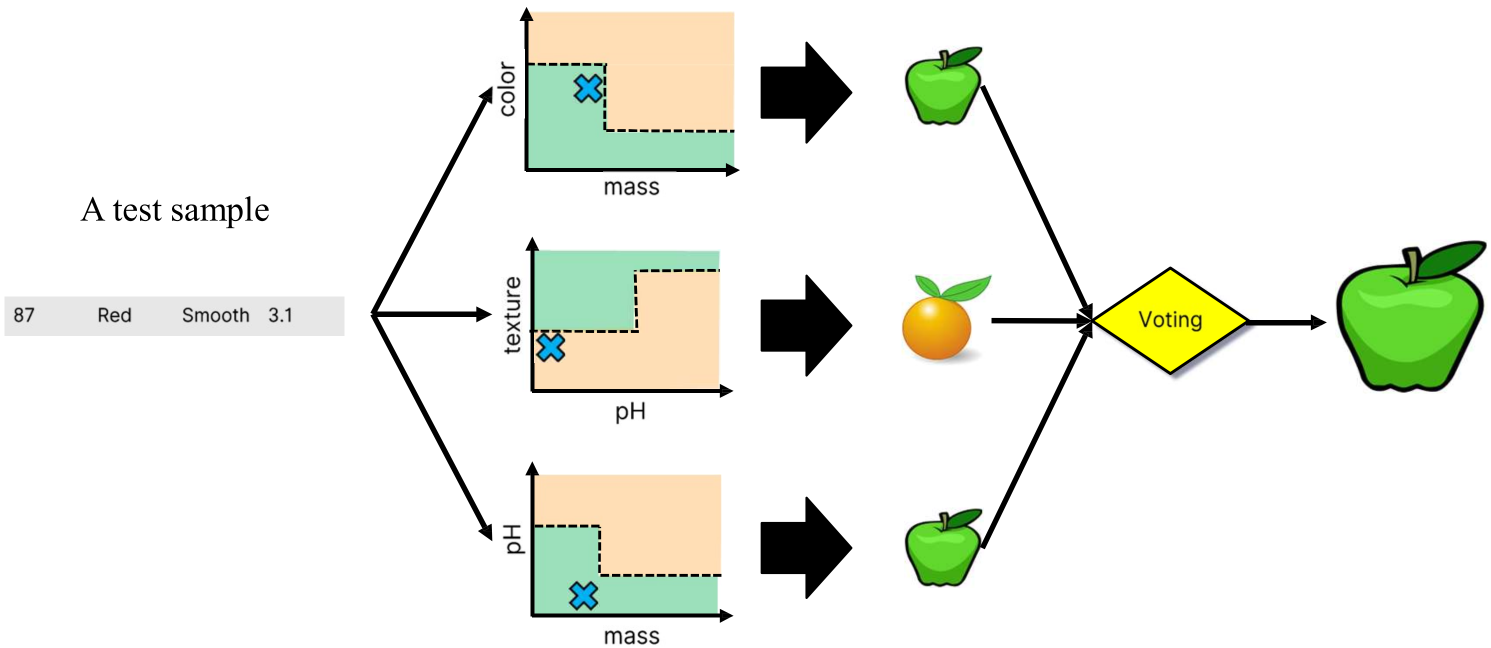
Random Subspace Method (Feature Bagging) at training time

Training data

Mass (g)	Color	Texture	pH	Label
84	Green	Smooth	3.5	Apple
121	Orange	Rough	3.9	Orange
85	Red	Smooth	3.3	Apple
101	Orange	Smooth	3.7	Orange
111	Green	Rough	3.5	Apple
...				
117	Red	Rough	3.4	Orange



Random Subspace Method (Feature Bagging) at inference time

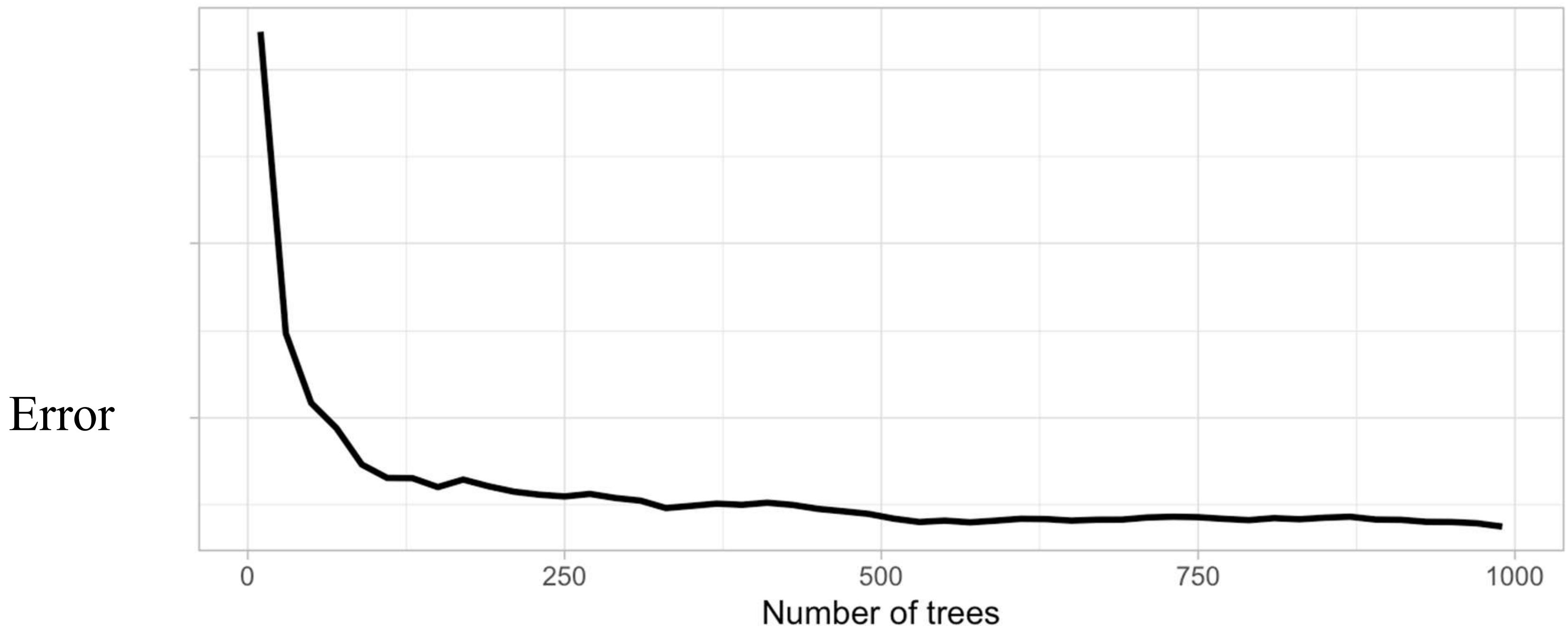


Impact of Depth and Number of Trees in Random Forests

- **Tree Depth:** Deeper trees can capture more complex relationships in the data. But excessively deep trees can lead to overfitting

- **Number of Trees:** More trees generally lead to better accuracy (low variance). However, too many trees can make the improvement become negligible, and computational cost increases.

Number of trees



Typical values

- **Default (in scikit-learn):** `n_estimators=100`
- **Common range:** 100–1000 trees
- **For larger datasets or more stable models:** sometimes up to 5000+

Random Forest (RF)

Advantages

- **More accurate** than a single decision tree. Standard decision trees often have high variance and low bias with High chance of overfitting (with 'deep trees', many nodes)
- With a RF, the bias remains low and the variance is reduced thus, we decrease the chances of overfitting. **Less overfitting** (because of averaging many trees)
- Works well with both **classification** and **regression**
- Handles **missing data** and **nonlinear relationships**

Disadvantages

- Slower to train and predict than a single tree
- Large models can require more memory